

M50: Final Composite Dependent Variable

PCA on Raw Switch Counts (Time, Topic-LDA, Typing)

No standardization (z-score) applied

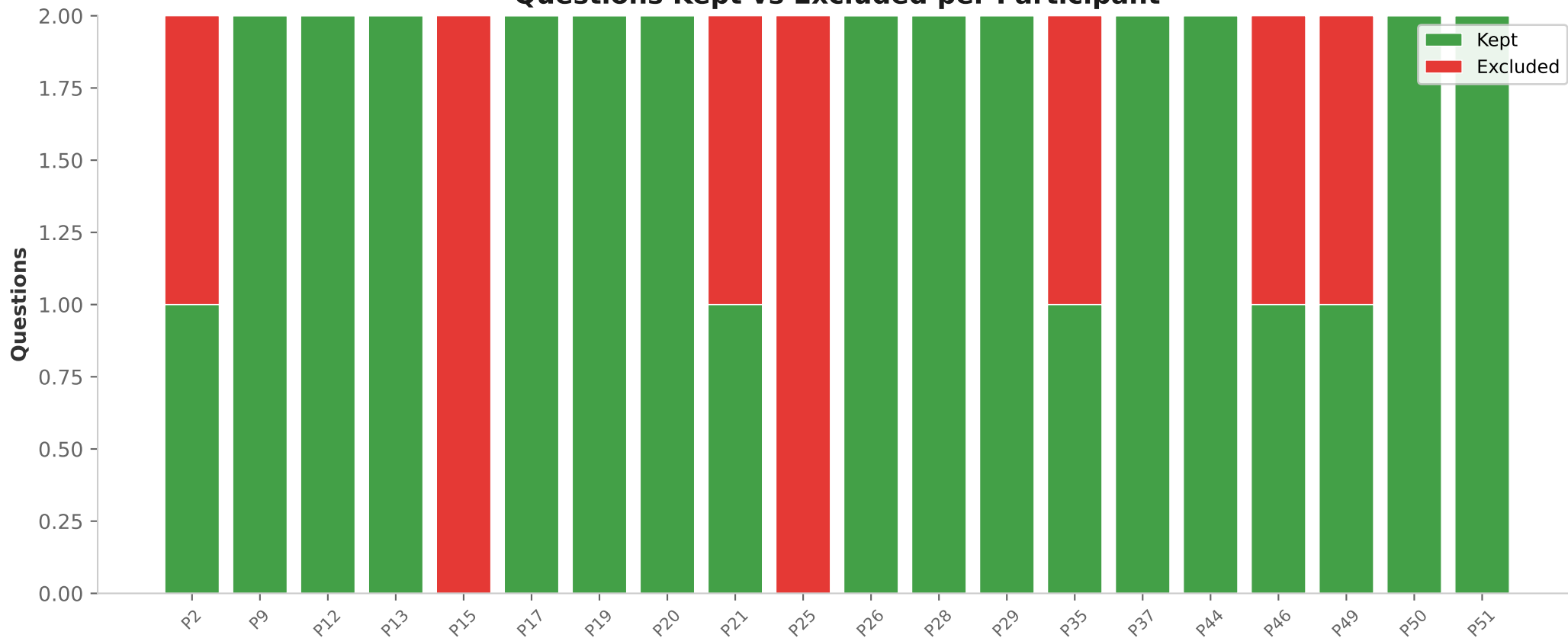
Exclusion Criteria Applied:

1. Questions with fewer than 3 page visits: 7 excluded
2. Questions with idle time $\geq 50\%$ after last meaningful event: 4 excluded
3. Participants > 3 SD from mean on any DV: 0 excluded

Result: 31 / 40 questions retained

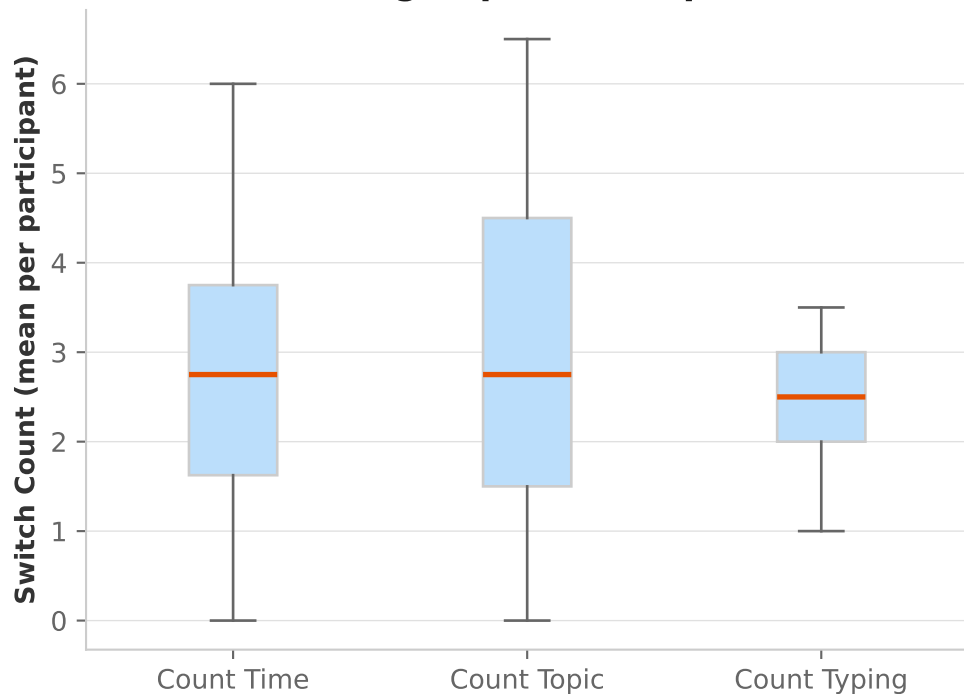
Final sample: N = 18 participants

Exclusion Summary: Questions Kept vs Excluded per Participant

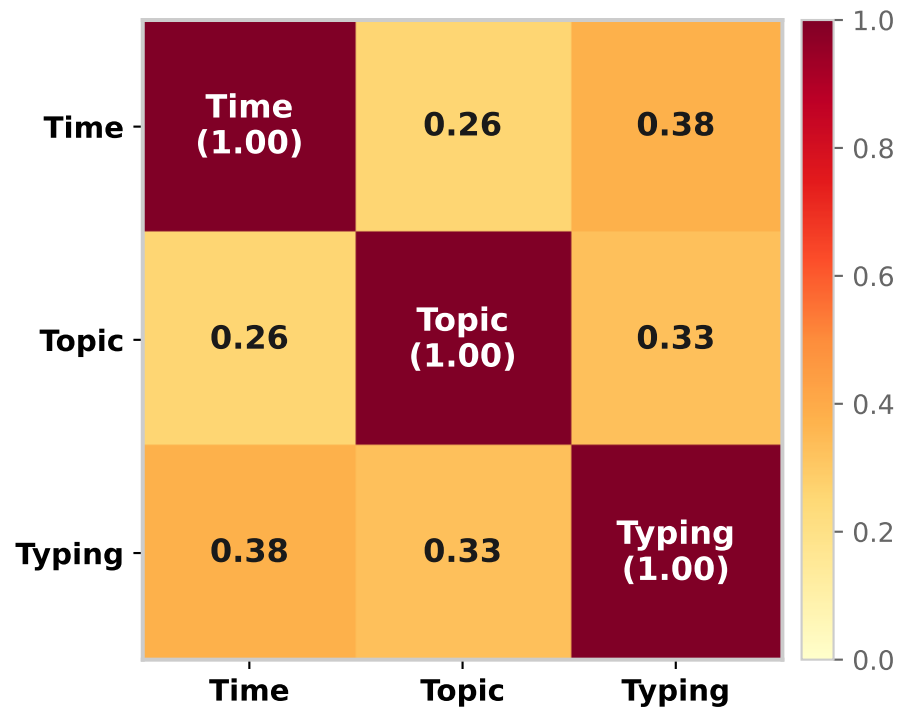


Descriptive Statistics: Raw Switch Counts

Distribution of Raw Switch Counts (Averaged per Participant)

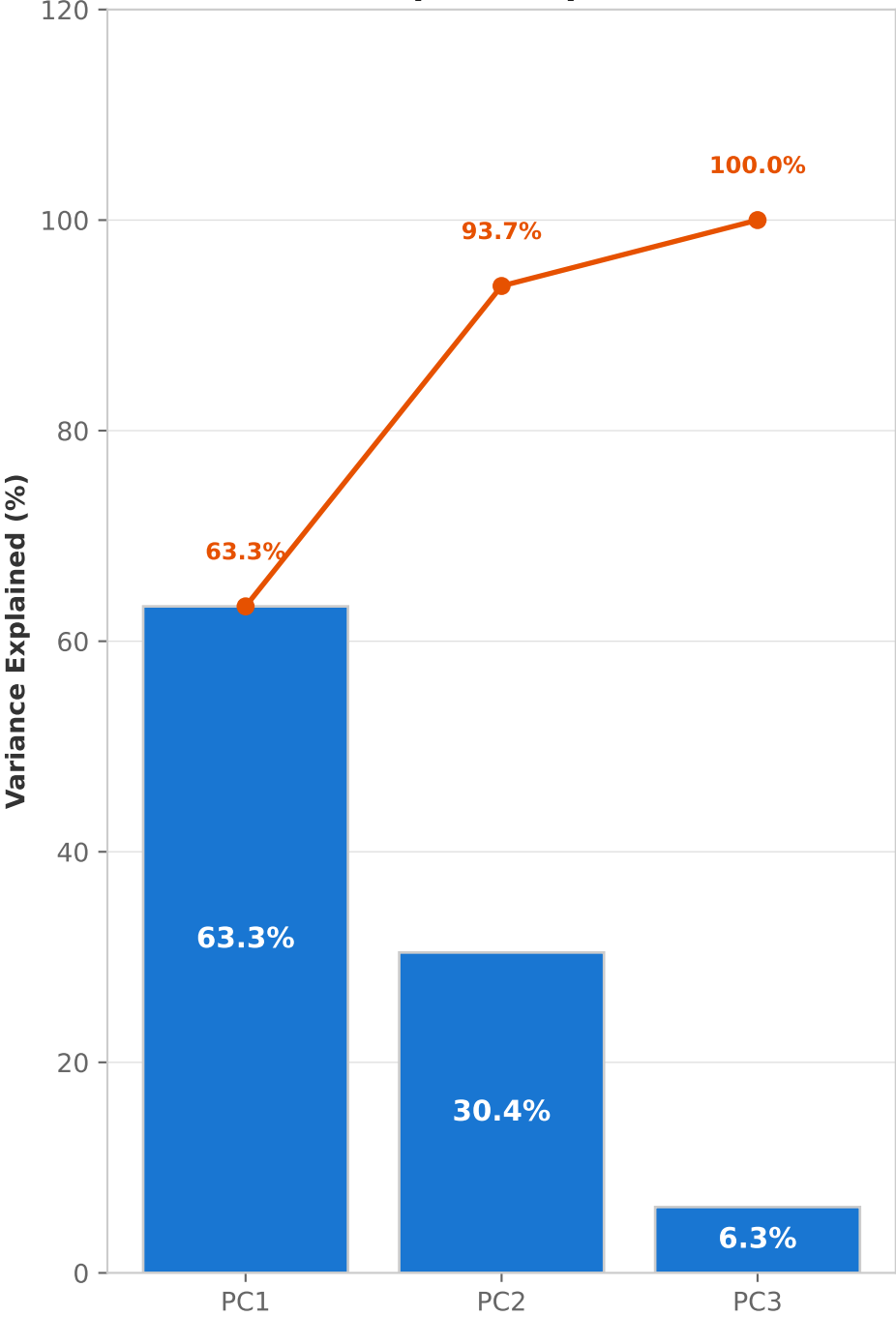


Correlation Between Switch Count Variables



PCA Results: Dimensionality Reduction on Raw Switch Counts

Variance Explained by Each Principal Component



PCA Loadings (variable weights per component):

PC1 (63.3% variance):			
Count Topic	+0.900	+++++	
Count Time	+0.404	+++++	
Count Typing	+0.162	+++	
PC2 (30.4% variance):			
Count Time	+0.897	+++++	
Count Topic	-0.425	-----	
Count Typing	+0.121	++	
PC3 (6.3% variance):			
Count Typing	+0.979	+++++	
Count Time	-0.177	---	
Count Topic	-0.097	-	

Interpretation:

- PC1: Primarily reflects topic switching (loading = +0.900)
Higher score = more topic transitions
- PC2: Primarily reflects time switching (loading = +0.897)
Higher score = more time transitions
- PC3: Primarily reflects typing switching (loading = +0.979)
Higher score = more typing transitions

Individual PC1 Scores: Composite Explore-Exploit Index

